# Performance Evaluation of Heuristic Strategies in Chain Reaction Game

## 1 Introduction

Chain Reaction is a turn-based strategy game where players add orbs to a grid, causing chain reactions and converting enemy cells. The game was developed on a 9-by-6 board, where two players—RED (human) and BLUE (AI)—take turns placing orbs. The AI determines its moves using the minimax strategy enhanced by alpha-beta pruning, guided by one of five distinct heuristic functions. To evaluate the AI's effectiveness under various strategies, an experiment was conducted using these heuristics as the basis for comparison.

## 2 Heuristic Defination

- Heuristic 1: comparing total orb counts.
- Heuristic 2:Rewards player's near-explosions and penalizes opponent's.
- Heuristic 3: Only penalizes opponent's near-explosions,fully defensive.
- Heuristic 4: Compares the number of occupied cells.
- Heuristic 5: Rewards player's cells that are not adjacent to opponent's cells, valuing safer, less vulnerable positions..
- Heuristic 6: Combined Heuristic (Weighted Sum of 1-5)

## 3 SETUP

The experiments were conducted with carefully defined configurations. For the human vs. AI setup, the RED player (human) made moves via mouse clicks, while the BLUE player (AI) used a minimax algorithm guided by one of five heuristics, with the search depth fixed at 2. Between five and seven games were played per heuristic to balance computational load and decision quality. In the AI vs. Random AI setup, BLUE employed minimax at a search depth of 3 using each of the five heuristics, whereas RED made purely random moves without search; one game per heuristic was played, totaling five games. For the AI vs. AI setup, both BLUE and RED utilized minimax with different heuristic combinations at a fixed depth of 2, with one game per heuristic pair, totaling ten games. No explicit time limits per move were imposed in any configuration. But there is a limit on moves of AI vs AI .

# 4 Results

## 4.1 Human vs. AI Results

The outcomes of the games are summarized in Table 1, showing the number of wins for each player per heuristic.

Table 1: Heuristic Based Comparison (Depth = 2)

| Heuristic | RED Wins (Human) | BLUE Wins (AI) |
|---|---|---|
| Heuristic 1 | 0 | 1 |
| Heuristic 2 | 0 | 1 |
| Heuristic 3 | 0 | 1 |
| Heuristic 4 | 0 | 1 |
| Heuristic 5 | 0 | 1 |

## 4.2 AI vs. AI Results

Table 2 shows the outcomes of the AI vs. AI games, with BLUE and RED using different heuristics.

Table 2: AI vs. AI Game Outcomes (Depth = 2)between 5 games

| Heuristic Pair (RED,BLUE) | RED Wins | BLUE Wins | WinRate(RED) | WinRate(Blue) |
|---|---|---|---|---|
| (1,1) | 5 | 0 | 100 | 0 |
| (1,2) | 0 | 5 | 0 | 100 |
| (1,3) | 5 | 0 | 100 | 0 |
| (1,4) | 0 | 5 | 0 | 100 |
| (1,5) | 5 | 0 | 100 | 0 |
| (1,random) | 5 | 0 | 100 | 0 |
| (2,1) | 5 | 0 | 100 | 0 |
| (2,2) | 5 | 0 | 100 | 0 |
| (2,3) | 5 | 0 | 100 | 0 |
| (2,4) | 5 | 0 | 100 | 0 |
| (2,5) | 5 | 0 | 100 | 0 |
| (2,random) | 5 | 0 | 100 | 0 |
| (3,1) | 0 | 5 | 0 | 100 |
| (3,2) | 0 | 5 | 0 | 100 |
| (3,3) | 5 | 0 | 100 | 0 |
| (3,4) | 0 | 5 | 0 | 100 |
| (3,5) | 5 | 0 | 100 | 0 |
| (3,random) | 4 | 1 | 75 | 25 |
| (random,random) | 3 | 2 | 60 | 40 |

### 4.3 AI vs. Random AI Results

Table 3 shows the outcomes of the AI (BLUE) vs. Random AI (RED) games, with BLUE using different heuristics.

Table 3: AI vs. Random AI Game Outcomes (Depth = 2)

| Heuristic (BLUE) | BLUE Wins | RED Wins |
|---|---|---|
| Heuristic 1 | 1 | 0 |
| Heuristic 2 | 1 | 0 |
| Heuristic 3 | 1 | 0 |
| Heuristic 4 | 1 | 0 |
| Heuristic 5 | 1 | 0 |

## 5 Discussion

### 5.1 Analysis

**Human vs. AI**: presents the results of Human vs. AI matches for five different heuristics at depth 2. For Heuristics 1 through 5, the AI (BLUE) won all games, while the human player (RED) failed to win any. Each heuristic resulted in 0 wins for RED and 1 win for BLUE, indicating a consistent 100% win rate for the AI across all tested heuristics.

**AI vs. AI**:In this study, five different heuristics were implemented for AI decision-making in the Chain Reaction game: (1) *Count Difference*, which evaluates the net difference in orb count between the player and the opponent; (2) *Near Explosion*, which rewards control over cells close to critical mass; (3) *Opponent Block*, which penalizes the opponent's potentially explosive cells; (4) *Domination*, which counts the number of cells controlled by each player; and (5) *Safe Zone*, which favors cells not adjacent to opponent-controlled cells. A random heuristic was also included for comparison. In AI vs. AI matches (each played 5 times per pairing at depth 2), heuristic 2 showed the most dominant performance, winning all matches regardless of the opponent's strategy. Heuristics 1, 3, and 5 performed well in specific pairings, often dominating weaker strategies. Notably, heuristic 4 consistently lost against other heuristics, indicating its limited strategic value. The random heuristic performed moderately, even winning 2 out of 5 games in the random vs. random setting. These results highlight the effectiveness of heuristics that consider explosion dynamics and safety in the game's tactical landscape. The test results from `output.txt` detail the performance of four heuristic configurations (`case1`, `case2`, `case3`, and `random`) in a two-player game across 16 combinations, with each combination consisting of 10 games. The configurations are defined by their heuristic weights as follows:

- **Case1**: $h_1 = 0.5$, $h_2 = 0.3$, $h_3 = 0.1$, $h_4 = 0.05$, $h_5 = 0.05$
- **Case2**: $h_1 = 0.1$, $h_2 = 0.7$, $h_3 = 0.1$, $h_4 = 0.05$, $h_5 = 0.05$
- **Case3**: $h_1 = 0.2$, $h_2 = 0.2$, $h_3 = 0.2$, $h_4 = 0.2$, $h_5 = 0.2$
- **Random**: No specific heuristic weights (assumed random moves)

Both `case2` and `case3` achieved the highest performance with 50 wins each, outperforming `case1` (40 wins) and `random` (10 wins). To differentiate between `case2` and `case3`, we consider their head-to-head performance:

- Combination 10: `case3` (Red) vs. `case2` (Blue) → `case2` wins 10, `case3` wins 0
- Combination 7: `case2` (Red) vs. `case3` (Blue) → `case3` wins 10, `case2` wins 0

The head-to-head results are inconclusive, as each configuration wins 10 games against the other. However, `case2` demonstrates greater consistency, as it does not lose to itself (Combination 6: 10 wins as Red), whereas `case3` loses to itself as Red (Combination 11: 0 wins).

**AI vs. Random AI**:In the AI vs. Random AI setup, the objective was to evaluate the performance of each heuristic-based AI against an opponent that selects moves without strategic planning. The RED player (AI) utilized the minimax algorithm with alpha-beta pruning and operated at a fixed search depth of 2. In contrast, the RED player (Random AI) made completely random legal moves, providing a consistent but unchallenging baseline for comparison. Each of the five heuristics was tested in a single game, resulting in a total of five matches. This configuration allowed us to observe how effectively each heuristic could dominate a non-strategic opponent, thereby establishing a minimum performance benchmark for each evaluation strategy. No time limits were imposed during these games to ensure decisions were made purely based on logic rather than speed.

## 5.2 Best Performing Heuristic

Based on the analysis of the `report.txt` file, which details 36 test combinations of heuristic configurations (case1, case2, case3, case4, case5, and random) in a two-player game with 5 games per combination, **case2** (`h1: 0, h2: 1, h3: 0, h4: 0, h5: 0`) emerges as the best heuristic, achieving the highest total wins of 55 out of 55 games (excluding the self-matchup in Combination 8).

Comprehensive comparisons show that case2 dominates by winning all games against case1 (Combinations 2, 7), case3 (Combinations 9, 14), case4 (Combinations 10, 20), and random (Combinations 12, 32), and only loses to case5 (Combination 26).

**Case1** (`h1: 1, h2: 0, h3: 0, h4: 0, h5: 0`) secures wins, performing strongly against case3 (Combinations 3, 13), case5 (Combinations 5, 25), and random (Combinations 6, 31), but loses to case2 (Combinations 2, 7) and case4 (Combinations 4).

**Case5** (`h1: 0, h2: 0, h3: 0, h4: 0, h5: 1`) follows with wins, excelling against case2 (Combination 26), case3 (Combinations 27), case4 (Combinations 28), and random (Combinations 30, 35), but losing to case1 (Combinations 5, 25).

**Case4** (`h1: 0, h2: 0, h3: 0, h4: 1, h5: 0`) achieves wins, beating case1 (Combinations 4), case3 (Combinations 16, 21), and random (Combinations 24, 34), but losing to case2 (Combinations 10, 20) and case5 (Combinations 28).

**Case3** (`h1: 0, h2: 0, h3: 1, h4: 0, h5: 0`) performs poorly with wins, only succeeding against case5 (Combination 27) and random (Combinations 18, 33), and losing to case1 (Combinations 3, 13), case2 (Combinations 9, 14), and case4 (Combinations 16, 21).

**Random** is the weakest, with only 10 wins across all matchups, gaining wins solely against itself (Combination 36: 3 wins as RED, 2 as BLUE) and a few scattered wins against case3 and case5 (1 win each in Combinations 18 and 30, respectively).

The dominance of case2 suggests that the feature evaluated by `h2` is critical to success in this game, making it the optimal heuristic configuration.